Chapter 1

Introduction

Exactly solvable (ES) potentials in quantum mechanics (QM) are very significant because they provide explicit solutions to the associated Schrödinger equation, which gives all important information and deep understanding of the physical systems. These explicit solutions obtained from ES potentials can be used to calculate various physical quantities such as transition probabilities, expectation values, scattering cross sections, etc. These potentials are important not only for theoretical studies but also for experimental physics as well. Further, these ES potentials serve as benchmark models that help in testing the accuracy and validity of various approximation methods. One of the remarkable properties of ES potentials is their ability to approximate more complex, non-ES systems. This is often achieved through perturbation theory or variational methods, where the ES solutions provide a useful starting point. However, the number of such ES potentials is quite limited, for instance, simple harmonic oscillator potential, Coulomb potential, Morse potential, Scarf II, Pöschl-Teller and Eckart potentials, etc. [1]. Owing to their significance, researchers made a lot of effort to construct new ES potentials. There has also been a resurgence of interest in finding exact solutions in QM, driven by the discovery of a new type of spectral problem called quasi-exactly solvable (QES) problems [2]. The specialty of these problems is that a limited number of initial eigenstates can be determined explicitly through algebraic methods. The concept of quasi-exact solvability refers to what the limits of exact (algebraic) knowledge can be at any given time. This means that the initial part of the spectrum is obtained not the full spectrum. The remaining special states are complex and can only be approximated. This discovery has further clarified the definition of an exact solution and provided a framework for identifying solvable problems that allow such solutions. Supersymmetric quantum mechanics (SUSY QM), shape-invariant potentials (SIPs), and algebraic methods like the factorization method, etc. are some of the approaches that are used to discover new ES potentials. These approaches often extend classical techniques or introduce new mathematical tools to discover and classify new ES potentials. For instance, SUSY QM provides a framework for constructing new ES potentials by relating them to known ones through SUSY transformations.

The concept of SUSY was first discovered in 1971 by Gel'fand and Likhtman [3], Ramond [4], and Neveu and Schwartz [5], and was later rediscovered by multiple research groups [6–9]. In particle physics, SUSY establishes the relationship between bosonic and fermionic degrees of freedom. It is worth noting that SUSY was first studied in its simplest form (SUSY QM) by Witten [10] and Cooper and Freedman [11]. In a later paper, Witten introduced the Witten index, a topological index designed to study SUSY breaking [12]. The first path integral formulation of SUSY QM was presented by Salomonson and van Holten [13]. Subsequently, it was demonstrated that the tunneling rate through double-well barriers could be precisely determined using SUSY methods [14–17]. In 1983, Gendenshtein [18] introduced the concept of a SIPs within the framework of SUSY QM. The concepts of SUSY have inspired new ideas in other branches of physics such as inverse quantum scattering [19], quantum gravity [20, 21], quantum theory of entanglement [22, 23], quantum computing [24], optical crystal [25], nuclear physics , condensed matter physics , atomic, and mathematical physics [7, 26–39].

After studying various aspects of SUSY QM, it was realized that it [1, 40] provides an understanding of the factorization approach developed by Infeld and Hull [41]. It has been realized that the factorization technique and technique of SUSY QM, with the concept of SI, are both reformulations [42] of Riccati's concept. This method of SUSY QM was the initial attempt to classify potential problems that may be solved analytically. Over the time, an entire methodology based on SUSY was developed to address solvable potential problems and to discover new solvable and SIPs. The concepts of SUSY QM and SIPs have not only enhanced our understanding of ES potentials but have also significantly expanded the list of known ES potentials. Using SUSY, a number of ES one-dimensional periodic potentials and QES periodic potentials are discussed [1, 40, 43, 44]. Cooper et al. [45] study the relationship among SUSY, shape invariance (SI), and solvable potentials [46–49]. These principles have also been utilized to expand the families of QES [44, 50, 51] and conditionally exactly solvable (CES) [52–54] potentials in non-relativistic QM. Subsequently, a new class of SIPs has been found, with a scaling parameter [55]. These potentials and multi step SIPs [56] have been discussed along with their relationship to self-similar potentials and q-deformations [1, 57–59]. The relationship between SUSY QM and the soliton solutions of the KdV hierarchy was also discussed [60–63]. Furthermore, there have been developments in approximate techniques based on SUSY, such as SUSY-inspired WKB (SWKB)

[64, 65], δ-expansion method [66], $1/N$-expansion with SUSY QM [67–69] and SUSY variational methods [70–72]. It turns out that the SWKB approximation maintains the exact SUSY relationships between the scattering amplitudes and the energy eigenvalues of the partner potentials [73]. The energy eigenvalue spectrum has been determined for several non-shape invariant systems [74–77]. It has been shown that the SWKB method is better in many situations than the conventional WKB approximation. There have also been attempts to derive the bound state eigenfunctions within the SWKB formalism [78–80]. Within the framework of SUSY QM, a number of aspects of the Dirac equation have also been examined [81–83]. Specifically, it has been demonstrated that by using the concepts of SUSY and SI, any analytically solvable one-dimensional Schrödinger problem has a corresponding Dirac problem in 1+1 dimensions with scalar interaction that is also analytically solvable. The factorization approach from SUSY QM has been used to solve the time-independent Dirac-Weyl equation to obtain exact analytical solutions for the bound states of a graphene Dirac electron in magnetic fields with translational symmetry [84, 85]. SUSY transformations have been utilized to design complex optical crystals that are transparent and one-way reflectionless (unidirectionally invisible) with balanced gain and loss profiles [25]. SUSY and SI formulations in a position-dependent mass (PDM) framework have been discussed in [86], where it has been shown that each quantum system with effective mass in one dimension has a corresponding supersymmetric partner system. The partner system has the same position dependence of the mass but with a different potential function.

In QM, the concept of PDM has captured the interest of many researchers over the past thirty years. Unlike traditional quantum systems where particle mass is constant, PDM systems feature particles whose mass changes based on their position. The study of PDM systems began with O. von Roos, who introduced a generalized form of the kinetic energy operator [87]. PDMs, first explored in the context of condensed matter physics, has now become an active area of study in other branches of physics. There has been notable progress in the methods used for crystallographic growth. The PDM method has gained interest because of its ability to produce non-uniform semiconductor specimens with abrupt heterojunctions. PDM problems are widely studied [88–90] and appear in various fields, including condensed matter physics, semiconductor physics, quantum liquids and quantum dots and wells [88, 91–95]. Over the years, researchers have applied PDM concepts to understanding a wide range of phenomena and materials. They have been used in studies involving the inversion potential for NH_3 in density functional theory, properties of helium clusters, optical properties of spherical quantum dots, and neutrino mass oscillations. Additionally, PDM systems are important in the study of nuclei, impurities in crystals, and metal clusters. In a recent work [96], the SUSY approach has been extended to all orderings of PDM hermitian as well as non-hermitian Hamiltonians. In Hermitian cases, the SUSY operators are unambiguous and reduced to Ben Daniel and Duke ordering. While, for non-Hermitian Hamiltonian, the PDM problems utilize the quasi-Hermitian nature, where each non-Hermitian Hamiltonian is connected to a Hermitian Hamiltonian by a similarity transformation. For these reasons, PDM problems have attracted significant interest in the literature from both quantum solvability and SUSY perspectives [97–102].

The main challenge in studying PDM systems is solving the Schrödinger equation as the mass varies with position. This leads to complex equations that are more difficult to solve than those with a constant mass. Researchers use various methods to tackle these problems, including coordinate transformations and SUSY techniques. Recent advancements include using the generalized von Roos Hamiltonian and point canonical transformations (PCT), which have provided exact solutions for certain PDM systems [103–106]. One significant advancement is the work by R. N. Costa Filho and his colleagues [107], who proposed a new way to handle PDM problems using a generalized translation operator. This method allows for tiny nonlinear displacements, providing new insights into PDM systems. PDM systems are also studied in quantum field theory by introducing a scalar potential in the mass term of the Klein-Gordon equation, allowing for the exploration of quark-antiquark interactions [108].

Recent research has further expanded the understanding and application of these systems. For example, Permatahati et al. [109] studied the Klein-Gordon equation with PDM in a Kratzer potential to determine the energy levels of particles. Yu et al. [110] investigated solutions to the PDM Schrödinger equation with the Morse potential using a series expansion method, highlighting the role of PDM in quantum dots. Another study by Maike et al. [111] explored the probability density correlation for PDM-Hamiltonians and interpreted the quantum partition function of the canonical ensemble of a PDM system in the context of superstatistics. Wibawa et al. studied the Klein-Gordon equation in the cosmic string and rainbow gravity spacetime for PDM particles [112]. Lima et al. studied the singular PDM [113]. Suparmi et al. studied the Dirac equation with the PDM for modified manning Rosen potential [114]. C Cari et al. studied the Dirac equation with the PDM for the modified woods saxon potential [115]. Yahiaoui et al. [105] used the group theoretic approach for Natanzon Potentials in PDM framework.

In physics, group theory finds broad applications. Physicists have recognized that dynamical groups, which include operators that do not commute with the Hamiltonian, can be more advantageous than symmetry groups from a computational perspective. [116, 117]. Using dynamical groups, it is possible to have transition from any eigenstate to another by applying a sequence of elements from the Lie algebra. This capability allows for the computation of all excited states from the ground state, which itself can be determined using variational methods. Thus, dynamical groups are valuable for calculating transition matrix elements and generating energy spectra. Using a group theoretic approach, Alhassid et al. [118–120] were able to determine the bound and the scattering states for one-dimensional Poschl-Teller and Morse potentials. Moreover, it was demonstrated that the scattering eigenstates constitute a basis for a specific representation of the non-compact symmetry group $SU(1,1)$. Subsequently, a purely algebraic procedure [119, 121] is introduced to derive recursion relations for scattering matrices (S-matrices) associated with systems belonging to this group. There are two variations of the $SU(1;1)$ or $SO(2;1)$ group [118, 119, 122] that differ from one another in terms of the role played by their Casimir invariants in physical contexts. The first version, known as the scattering state group, is characterized by a Casimir invariant whose eigenvalues determine the strength of the potential. The representation of this group includes scattering states with different energy levels for a constant potential. The second version, known as the potential group, also has a Casimir invariant, but its representation includes fixed energy states with varying strengths. Subsequently, the Euclidean connection and the construction of S-matrices for potential symmetry groups $SO(2,1)$, $SO(2,2)$, and $SO(2,3)$ in one, two, and three dimensions, respectively, were also derived [123]. A proposal has been put forward [124] to realize the $SO(2,2)$ group within a family of solvable Natanzon potentials [125, 126]. This family includes the well-known Pöschl-Teller, Morse, Rosen-Morse, Manning-Rosen, and Eckart potentials. The bound and scattering state spectra of these potentials are obtained using a group theoretic approach. A thorough investigation has been carried out to identify $SO(2,1)$ algebraic structures related to SIPs, utilizing a specific differential realization of the generators [127]. The group theoretic approach is also useful for studying the relativistic Coulomb-Dirac scattering [128] in external fields. Furthermore, the potential groups $SO(2,2)$ and $SO(2,1)$ are developed in Ref. [129] to describe the hypergeometric and confluent hypergeometric equations, respectively.

The bound state solutions associated with ES systems are always in terms of hypergeometric or confluent hypergeometric functions, which can be further expressed in terms of one of the well-known classical orthogonal polynomials (COPs) such as Jacobi, Laguerre, Hermite, etc. The significance and applications of these COPs have been extensively investigated, and they are widely used in many branches of physics. All these sets of COPs have many properties in common. Each constitutes a set of polynomials of increasing degree starting from $n = 0$. They satisfy the orthogonality condition over a certain interval and with a certain non-negative weight function and hence constitute an orthonormal basis for expansion of certain functions over the appropriate interval. Each of these COPs satisfy a second-order linear homogeneous differential equation, and hence these are used in the study of many physical problems. Each of these COPs is generated through some generating functions and satisfy the three-term recursion relation. However, the number of such sets of COPs is only a few, and this is probably the reason behind the limited number of ES potentials in QM.

In the year 2009, an important breakthrough occurred in the field of orthogonal polynomials. Three mathematicians, David Gomez-Ullate, Niky Kamran, and Robert Milson, are able to find new families of orthogonal polynomials related to some of the COPs [130]. Unlike usual COPs, these sets of polynomials start from the degree $n > 0$ and still form a complete set of orthonormal basis with respect to a different positive definite weight function. These new sets of polynomials that also satisfy the Strum-Liouville problem are known as exceptional orthogonal polynomials (EOPs). Only two COPs, namely Laguerre polynomials and Jacobi polynomials, are found to have their exceptional extension. When $n = 1$, i.e., the set of polynomials starts with degree one, these polynomials are known as X_1-EOPs, namely X_1-Laguerre or X_1-Jacobi polynomials. If the set of polynomials starts with an arbitrary positive integer m, i.e., $n = m$, these polynomials are known as X_m EOPs. In subsequent studies, people have found quantum mechanical systems whose bound state solutions are written in terms of EOPs. Hence the discovery of EOPs has enlarged the list of ES systems in QM. The potential associated with these new ES systems are known as rationally extended potentials. C. Quesne constructed [131] two rationally extended potentials corresponding to conventional radial oscillator potential and Scarf-I potential using PCT approach and wavefunctions of these extended potentials are written in terms of X_1-Laguerre and X_1-Jacobi EOPs respectively. Bagchi et al. [132] used the SUSY approach and constructed the rationally extended generalized Pöschl-Teller (GPT) potentials, rationally extended Scarf I potential, and ratinally extended PT- symmetric Scarf II potential. Yves Grandati [133] provide a new method to construct rationally isotonic oscillator assosiated with Laguerre EOPs. Choon-Lin Ho [134] used the prepotential approach and find the rationally extended radial oscillator potential. Afterthat, By using SUSYQM, Marquette et. al. [135] constructed the rationally extended Harmonic oscillator potential associated with Hermite EOPs. Various properties assosiated with these rationally extended potentials have been discussed in Refs. [136–148].

In the early 1990s, researchers investigated some examples of Hermite EOPs [149], which did not recieve any attention. The systematic study on EOPs starts with two papers [130, 150], where they give a full classification of X_1 polynomials (codimension one i.e. total number of missing degrees). The discovery of EOPs has provided new opportunities for research, especially in the study of solvable potentials where the solution of the associated Schrodinger equation may be expressed in terms of these EOPs. Quesne et al. [131, 132] reformulated this new result in field of QM and observed the connection between EOPs and SIPs. She was the first to identify the significance of Darboux transformations in constructing EOPs and introduced the first two codimension examples (X_2-Laguerre and X_2-Jacobi) [151]. Odake et al. [152–154] were the first to form higher-codimension families. In their study, the authors applied the prepotential approach [134, 155] and obtained two sets of infinite SIPs and corresponding eigenfunctions in terms of X_m Laguerre and X_m Jacobi EOPs. Later, in Ref [156], equivalent but more straightforward forms of the Laguerre- and Jacobi-type X_m EOPs were introduced. These simplified forms make it easier to explore several key properties of the X_m EOPs. These properties include the effects of the forward and backward shift operators on the X_m polynomials, the Gram-Schmidt orthonormalization process used in their algebraic construction, as well as the Rodrigues formulas and generating functions associated with these new polynomials. The characteristics of these X_m EOPs, such as the Rodrigue formulae and recursion relations, have also been discussed in detail in Refs. [156–158]. Recently, these EOPs have been extensively studied from various perspectives. For example, potential applications of EOPs in PDM systems was explored in Ref. [136], while relevance of EOPs in Dirac and Fokker-Planck equations was examined in Ref. [138]. Additionally, these new polynomials have been investigated as solutions corresponding to certain CES potentials, as discussed in Ref. [159].

Afterthat, these EOPs are constructed by using Darboux-Crum transformation. Ullate et al. [160] characterized the X_m-Laguerre polynomials using an isospectral Darboux transformation, which was initially introduced in Refs. [161, 162]. They also demonstrated that the SI of these new polynomial families is due to the permutability property of the Darboux-Crum transformation. They provide an example of the higher codimension generalization of Jacobi polynomials [163] and conduct a whole analysis of the parameter values for which these polynomial families possess non-singular weights. The authors achieved this by utilizing the isospectral algebraic Darboux transformation. In another work Sasaki et al. [150, 164, 165] provided a better understanding of the Darboux transformations. The further development in the study of EOPs involved multi-step or higher-order Darboux transformations, leading to the creation of exceptional families known as multi-index EOPs [163, 166, 167]. Other equivalent methods to construct EOP families have been discussed in a number of works. These methods include the prepotential approach [155] and the use of symmetry groups to preserve the form of the Rayleigh Schrödinger equation [168]. These approaches result in the development of rational extensions of well-known solvable potentials. Exceptional polynomial systems are most often found as eigenfunctions to ES quantum mechanical systems, including both bound states [133, 151, 152, 159, 164, 169–172] and scattering amplitudes [140, 141, 173, 174]. But some superintegrable system are also connected to EOPs [175, 176]. These are also linked with higher-order symmetry [135, 176, 177], Dirac equations coupled with some external fields, Fokker-Planck equations [138, 178, 179], Shannon entropy for some CES potentials [139], semifinite-gap potentials [180] and Dirac equation involving scalar or pseudo-scalar potential. The study of exceptional Hermite polynomials, which are associated with a single-step state-adding

Darboux transformation, was initially examined in Ref. [181] and then discussed in Refs. [159, 182–184] and two-step Darboux transformations considered in Ref. [185]. Wronskian determinants for the Hermite polynomials family have been discussed in Ref. [186]. The classification of EOPs is discussed in Refs. [187, 188], the characteristics of their zeros in Refs. [158, 189, 190], and their recurrence relations are given in Refs. [191–194].

Recently, a new construction of multi-parameter exceptional Legendre polynomials was introduced by considering the isospectral deformation of the classical Legendre operator [195]. They used a confluent Darboux transformation to generate EOP families with an arbitrary number of real parameters. After discovering these EOPs, researchers expanded the list of ES potentials and obtained their bound state wavefunctions in terms of these EOPs [131, 134, 152–155]. It is worth noting that the recently discovered potentials are a rational extension of the conventional potentials; it means a rational term includes the usual potentials. Rationally extended generalized Poschl Teller, trigonometric Scarf, and radial oscillator potentials are included in this set. The scattering state solutions for some real and hermitian potentials were first derived by Yadav et al. [140, 173, 174]. The bound state spectrum remains the same, but these extended potentials' scattering amplitudes differ with an additional multiplicity k-dependent component. After that, Sasaki et al. [141] got the same for multi-index solutions using Jacobi and Laguerre polynomials along with EOPs as the main component of the eigenfunctions for the rationally extended potentials. Interestingly, these potentials are formulated within the framework of SUSY QM, and it has been shown that these potentials satisfy the SI condition. SUSY plays an important role in the study of newly discovered ES systems, with their solutions expressed in terms of EOPs.

Motivation

As mentioned earlier in this thesis, there are only a few potentials that are ES in QM. For example, the simple harmonic oscillator (SHO), the hydrogen atom (H-atom) problem, and a class of spherically symmetric potentials. The solutions to these systems are typically expressed in terms of classical polynomials. For instance, the solution to the SHO is given by Hermite polynomials, the H-atom by Laguerre polynomials, and spherically symmetric potentials generally by Legendre and associated Legendre polynomials. However, after the discovery of EOPs, there has been a significant expansion in the list of ES potentials. These new potentials are often rational extensions of conventional potentials and are referred to as rationally extended potentials. Various techniques have been employed to achieve rational extensions, including the PCT approach, SUSY techniques, and the prepotential approach [131–133, 135]. A unique feature of these potentials is that both the potential itself and the associated wavefunction differ entirely from the conventional potential and its wavefunction, yet the eigenvalues remain the same. This means they are strictly isospectral with the conventional potentials. It will be interesting to explore the reason behind this isospectrality.

In QM, the simplest case is the harmonic oscillator problem, which is associated with the Hermite equation and respective orthogonal polynomials. The solubility of this equation can be easily attributed to SI. Any system exhibiting spherical symmetry naturally leads to the Legendre and associated Legendre equations, with their standard solutions being the respective orthogonal polynomials. Ashok Das et al. [196] studied these equations from an operator point of view, similar to the harmonic oscillator, and showed that SI underlies their solvability. This work was later extended to the hypergeometric case [197], illustrating that the solubility of systems described by the hypergeometric equation and its associated functions is also a result of underlying SUSY and SI. Further, the above discussion raises two key questions simultaneously. First, why are conventional potentials and their rationally extended counterparts isospectral? What underlying factors contribute to their isospectrality? Second, can we study the differential equations of EOPs from an operator point of view? And do SUSY and SI also play a role in their solvability?

In recent years, significant interest has been attracted by quantum mechanical systems with PDM due to their relevance in describing the physics of many contemporary microstructures. The focus on the effective mass approach arises from the remarkable advancements in crystallographic growth techniques, enabling the production of non-uniform semiconductor specimens with abrupt heterojunctions. The PDM is used to develop pseudo-potentials, which provide a notable computational benefit in the quantum Monte Carlo approach [198, 199]. The wide range of applications accounts for the increasing interest in developing solvable cases of the effective mass Schrödinger equation. Several such cases have been obtained using Lie algebraic techniques or supersymmetric methods and the PCT approach [86, 100, 102, 103, 200–216]. Bagchi et al. [217] investigated the one-dimensional Schrodinger equation in a PDM framework. They obtain some new ES or QES potentials by using the PCT technique and assuming different types of PDM. The eigenfunctions corresponding to ES potentials are related to either Jacobi or generalized Laguerre polynomials, while the latter are related to certain non-hypergeometric polynomials of degree k. After discovery of EOPs, Midya et. al. [136] obtained the ES potentials by using the PCT approach in a PDM framework whose bound state solutions are in terms of X_1 Laguerre and Jacobi EOPs. These potentials are rational extensions of the ones found in Ref. [217]. They also showed that these new ES potentials satisfy the condition of SI and are isospectral to the previously obtained potentials in PDM background whose wavefunctions are written in terms of classical Jacobi and Laguerre polynomials.

Despite the numerous rational extensions, the number of ES potentials within the PDM framework remains quite limited. Therefore, discovering new ES potentials in the PDM framework, where the associated wavefunctions can be expressed in terms of EOPs, would be highly intriguing.

In QM, it is interesting to find zero modes because, contrary to common belief, these modes can be dynamic and important as any other energy states. In the study of quantum systems with nonconfining potentials, the focus is usually on bound states ($E < 0$) and free states ($E > 0$). However, zero-energy states (E = 0) have unique properties that often go unnoticed. While in some systems, like the Coulomb problem, the $E = 0$ state is non-normalizable and part of the continuum. There are notable exceptions where the $E = 0$ state is bound and normalizable for specific values of the coupling constant. One of the main challenges to finding zero-energy quantum states is obtaining the normalized solutions corresponding to the Schrödinger equation. This has encouraged many studies to determine the existence of these zero-energy states that possess real asymptotic properties [218–225]. These discoveries have significant theoretical and practical implications, challenging the assumption that all zero-energy states lie in the continuum. The existence of bound $E = 0$ states expands our understanding of quantum systems and provides new avenues for research. Daboul and Nieto [219] have systematically studied radial power law potentials and obtained some unusual solutions. The wavefunction corresponding to $E = 0$ is normalizable, bound, or unbound. It was observed [226] long ago, that there might be zero-energy solutions that exhibit both confinement (in the normalized case) and leakage (in the non-normalized case) for certain kinds of interacting systems. Bagchi et al. [227] provided another example of zero-energy normalizable solutions. They constructed QES potentials using $SO(2,1)$ potential algebra, resulting in wavefunctions that are normalizable in one case. Therefore, it is worthwhile to present other examples where zero-energy quantal states are normalizable. These examples can provide deep insights into the conditions under which zero-energy states become normalizable and can reveal new potential applications in QM. Based on these developments and motivations, we have attemped to investigate the issues related to EOPs and to construct some zero-energy quantum state corresponding to a many-body system in this thesis.

The main motivations in this thesis are to investigate these problems under the common umbrella of EOPs. Asok Das et al. have studied COPs [196] and hypergeometric functions [197] in the SUSY framework. We followed their approach to study the EOPs differential equations in the framework of SUSY QM. We first consider the differential equations associated with X_1-Jacobi and X_1-Laguerre EOPs [130, 228]. We treat these differential equations as eigenvalue equations and draw an analogy with the time-independent Schrödinger equation to define a Hamiltonian. This approach allows us to study these

Hamiltonians within the framework of SUSY and to explore the SI associated with such systems. We demonstrate that the underlying SI symmetry is responsible for the solubility of the differential equations associated with these EOPs [229]. We further extend our study to more general cases (X_m Jacobi and X_m Laguerre EOPs) and demonstrate that the solubility of differential equations associated with these EOPs is due to underlying SUSY and SI.

In our second work, we focus on constructing new ES potentials within a PDM background. We utilize the PCT approach within the PDM framework to achieve this. The PCT approach is known [230] to transform a Schrödinger equation with one potential into a Schrödinger equation with a new potential. The literature offers various choices for mass, which are position-dependent. We begin by using a well-known PDM [136] to construct ES potentials that depend on a parameter m. We then derive the eigenvalues and eigenfunctions for these potentials. The eigenfunctions corresponding to these potentials are expressed in terms of X_m Laguerre EOPs. For a particular value, i.e., $m = 1$ our study resemble with work of Midya et al. [231]. Next, we choose a new PDM that depends on a parameter ν [232]. This leads to a one-parameter family of solvable potentials. We then derive the eigenvalues and eigenfunctions for these potentials. The eigenfunctions of these new one parmeter (ν) family of potentials are expressed in terms of X_m Laguerre EOPs, which also depend on the parameter ν. Furthermore, we examine these two cases within the SUSY PDM framework and find that both potentials are shape invariant, which is the underlying symmetry ensuring their exact solvability.

Next in this thesis, we investigate the existence of regular zero-energy normalizable solutions for a system of QES potentials that corresponds to a rationally extended many-body truncated Calogero-Sutherland model. We address three cases corresponding to the restrictions on the guiding functions of $SO(2,1)$ [233]. A significant outcome of our analysis is that in all three cases, we find normalizable wavefunctions by imposing plausible restrictions on the coupling parameters [234]. This is in contrast to a similar situation in a previous work [227], where a class of QES potentials with known E = 0 was addressed. In that case, it was found that, subject to certain convergence conditions, the wave function of the zero-energy state is normalizable for only one class of potentials. In our work, we begin with a rationally extended truncated Calogero-Sutherland model. We apply the PCT approach to the associated Schrödinger equation and match the resulting energy expression with the Casimir of the $SO(2,1)$ potential algebra. This results in three new classes of QES rational potentials corresponding to the E = 0 state. For all of them, we demonstrate that the solutions for the eigenvalue equation at zero energy are normalizable and asymptotically smooth, provided the coupling parameters are appropriately restricted. Based on our research and the new findings, this thesis is organized into six chapters, as detailed below.

Plan of the Thesis

In **chapter 2**, we introduce essential mathematical tools that are foundational for the rest of the thesis. We start by EOPs, covering their differential equation, specific examples of X_1-Jaobi and X_1-Laguerre polynomials and the corresponding weight functions. Then, we generalize these differential equations for the more general cases, namely X_m-Jacobi and X_m-Laguerre polynomials. We further provide a brief overview of the techniques used in SUSY QM, including the construction of a hierarchy of Hamiltonians and SI potentials. After that, we briefly discuss the mathematical tools like the PCT approach and the group theoretic approach.

In **chapter 3**, we address the fundamental question, why rationally extended potentials are strictly isospectral to the usual potentials. We found the answer of this fundamental question by considering the X_1-Jacobi and X_1-Laguerre differential equations as eigenvalue equations to define Hamiltonians and study them within the SUSY framework. By constructing a hierarchy of Hamiltonians that exhibit SI, we determine the eigenvalues of the original Hamiltonian as we expected. We demonstrate that the solubility of these differential equations is due to the underlying SI symmetry. Subsequently, we extend our study to X_m-Jacobi and X_m-Laguerre polynomials, showing that the solubility of these differential equations is due to both SUSY and SI symmetries, which are the reason for isospectrality of rationally extended potentials for which the wavefunctions are written in terms of these EOPs.

In **chapter 4**, we start with the PCT approach in a PDM background to construct ES potentials. We solve the Schrödinger equation with two different PDM systems. In the first case, we identify the ES potential for which the eigenfunctions are expressed in terms of X_m-Laguerre polynomials. In the second case, we use a new PDM system that depends on some parameter ν. This results in a one-parameter (ν) family of potentials, with the corresponding eigenfunctions also expressed in terms of X_m-Laguerre EOPs. Following this, we use SUSY techniques to find the SIPs corresponding to the potentials in both cases. We find that the exact solvability of these potentials is due to SI symmetry.

In **chapter 5**, we focus on finding normalizable zero-energy quantum states. We examine a rationally extended truncated Calogero-Sutherland model by applying a PCT to the associated Schrödinger equation. We then match the resulting energy expression with the Casimir of the $SO(2,1)$ algebra, leading us to identify three classes of QES potentials. For each class, with suitable restrictions, we find normalizable wavefunctions that are asymptotically smooth.

In **chapter 6**, we summarize the results presented in the various chapters of this thesis and discuss future prospects for our research.